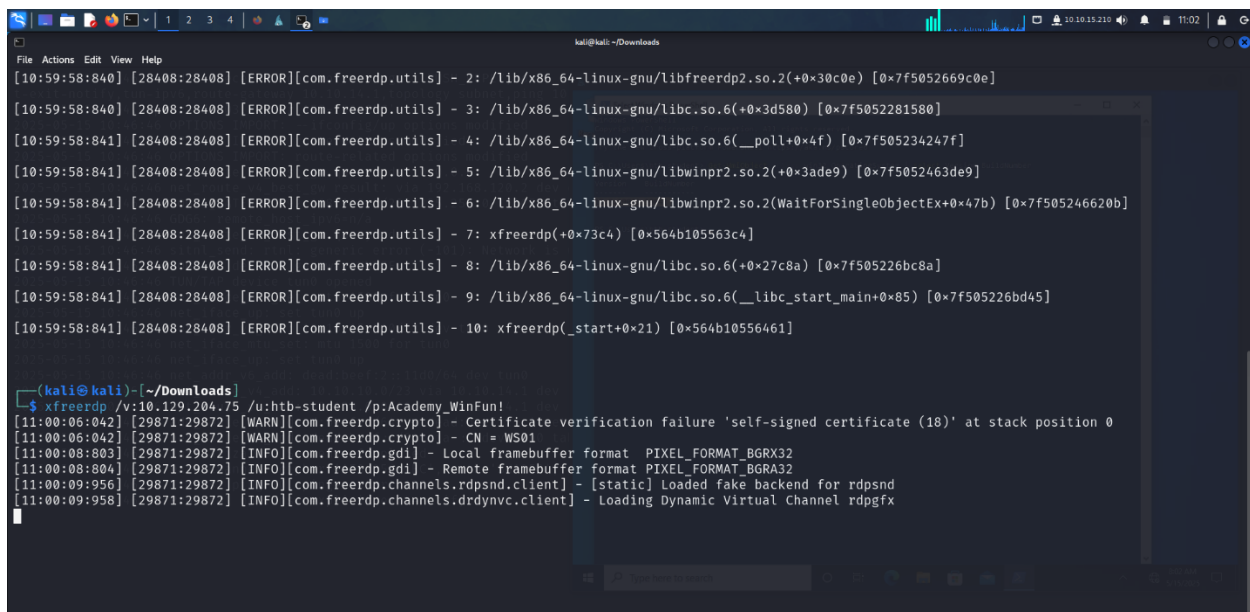


1. Introduction

Windows remains the dominant operating system in corporate environments, making it a critical focus for penetration testers and security professionals. Understanding how to interact with Windows systems is essential for effective enumeration, privilege escalation, lateral movement, and post-exploitation. This module introduces the foundational concepts of the Windows operating system and command line, equipping learners with the skills needed to operate on both target and attack machines during assessments.

2. Module Sections

- I connected to the machine via xfreerdp and ran the command as shown



```
kali@kali: ~/Downloads
File Actions Edit View Help
[10:59:58:840] [28408:28408] [ERROR][com.freerdp.utils] - 2: /lib/x86_64-linux-gnu/libfreerdp2.so.2(+0x30c0e) [0x7f5052669c0e]
[10:59:58:840] [28408:28408] [ERROR][com.freerdp.utils] - 3: /lib/x86_64-linux-gnu/libc.so.6(+0x3d580) [0x7f5052281580]
[10:59:58:841] [28408:28408] [ERROR][com.freerdp.utils] - 4: /lib/x86_64-linux-gnu/libc.so.6(__poll+0x4f) [0x7f505234247f]
[10:59:58:841] [28408:28408] [ERROR][com.freerdp.utils] - 5: /lib/x86_64-linux-gnu/libwinpr2.so.2(+0x3ade9) [0x7f5052463de9]
[10:59:58:841] [28408:28408] [ERROR][com.freerdp.utils] - 6: /lib/x86_64-linux-gnu/libwinpr2.so.2(WaitForSingleObjectEx+0x47b) [0x7f505246620b]
[10:59:58:841] [28408:28408] [ERROR][com.freerdp.utils] - 7: xfreerdp(+0x73c4) [0x564b105563c4]
[10:59:58:841] [28408:28408] [ERROR][com.freerdp.utils] - 8: /lib/x86_64-linux-gnu/libc.so.6(+0x27c8a) [0x7f505226bc8a]
[10:59:58:841] [28408:28408] [ERROR][com.freerdp.utils] - 9: /lib/x86_64-linux-gnu/libc.so.6(__libc_start_main+0x85) [0x7f505226bd45]
[10:59:58:841] [28408:28408] [ERROR][com.freerdp.utils] - 10: xfreerdp(_start+0x21) [0x564b10556461]

(kali@kali) - [~/Downloads]
$ xfreerdp /v:10.129.204.75 /u:htb-student /p:Academy_WinFun!
[11:00:06:042] [29871:29872] [WARN][com.freerdp.crypto] - Certificate verification failure 'self-signed certificate (18)' at stack position 0
[11:00:06:042] [29871:29872] [WARN][com.freerdp.crypto] - CN = WS01
[11:00:08:803] [29871:29872] [INFO][com.freerdp.gdi] - Local framebuffer format PIXEL_FORMAT_BGRX32
[11:00:08:804] [29871:29872] [INFO][com.freerdp.gdi] - Remote framebuffer format PIXEL_FORMAT_BGRA32
[11:00:09:956] [29871:29872] [INFO][com.freerdp.channels.rdpnd.client] - [static] Loaded fake backend for rdpnd
[11:00:09:958] [29871:29872] [INFO][com.freerdp.channels.drdynvc.client] - Loading Dynamic Virtual Channel rdpgfx
```

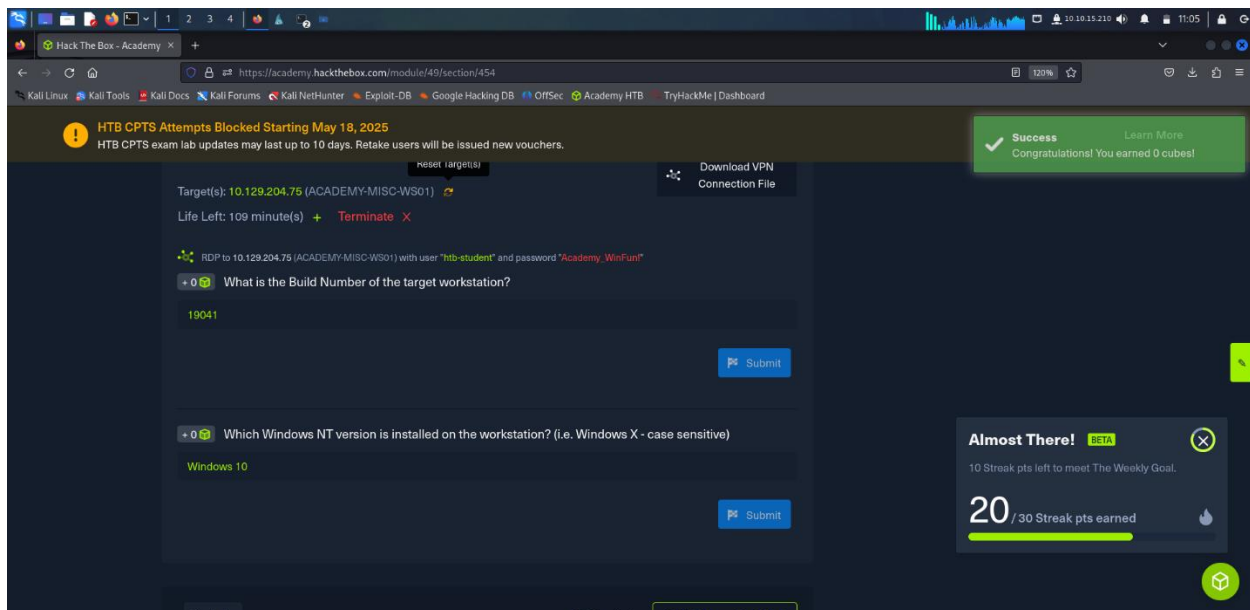
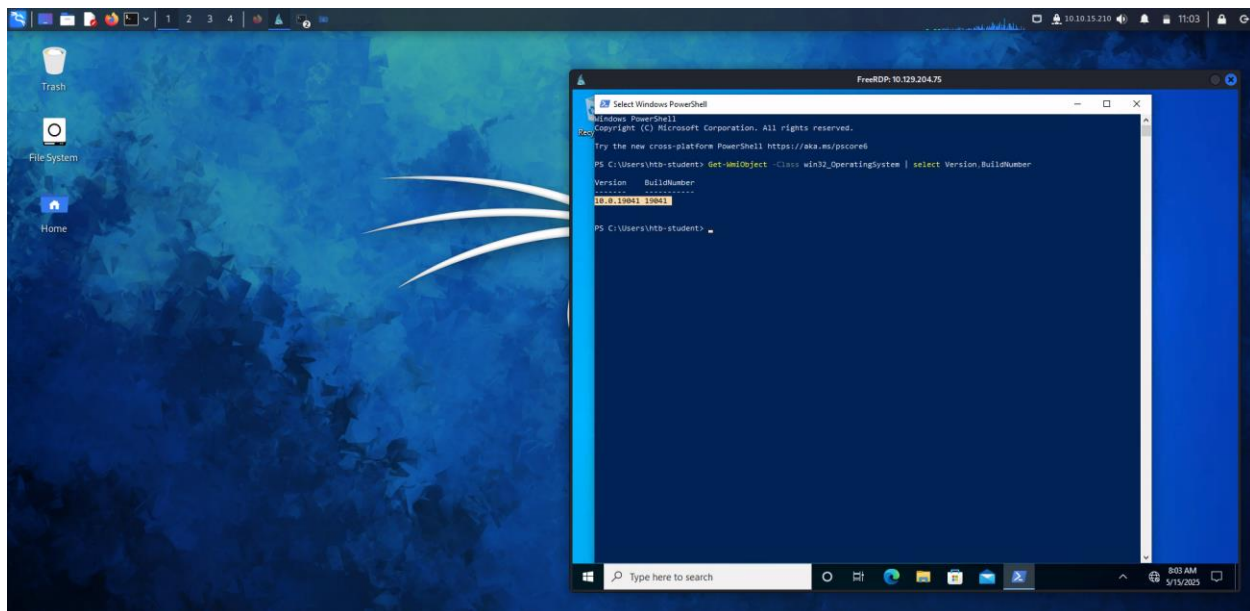
I then ran the command shown below to get the Windows version installed and the build number.

Q What is the build number of the target workstations

Answ 19041

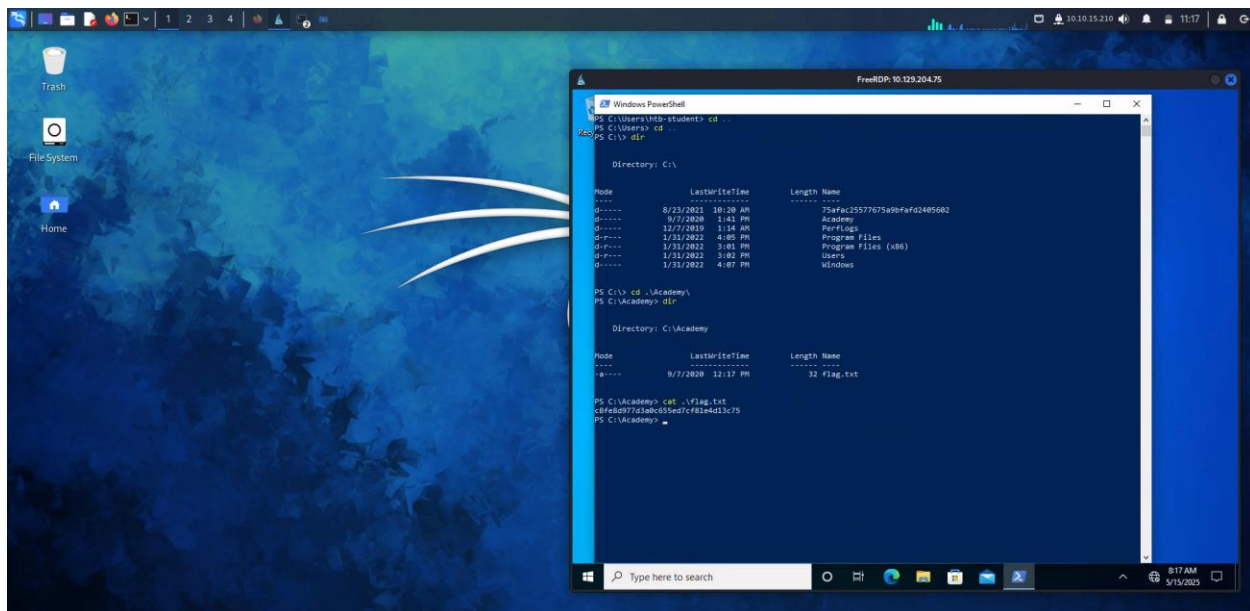
Q Which Windows NT Version is installed in the workstation?

Answ Windows 10



Question: Find the non-standard directory in the C:\. Submit the contents of the flag file saved in this directory. Answer: ce8f8ed977d33c06b5e7cf814e413c75

I listed the directories in the directory C using dir command. I then changed directories and went to Academy and found the flag file and displayed the content using cat command



HTB CPTS Attempts Blocked Starting May 18, 2025
HTB CPTS exam lab updates may last up to 10 days. Retake users will be issued new vouchers.

Success
Congratulations! You earned 0 cubes!

Learn More

Questions

Answer the question(s) below to complete this Section and earn cubes!

Target(s): 10.129.204.75 (ACADEMY-MISC-WS01) 🔄

Life Left: 98 minute(s) + Terminate ✕

🔑 RDP to 10.129.204.75 (ACADEMY-MISC-WS01) with user "htb-student" and password "Academy_WinFun!"

🔑 Find the non-standard directory in the C drive. Submit the contents of the flag file saved in this directory.

c8fe8d977d3a0c655ed7cf81e4d13c75

Submit Hint

Cheat Sheet

Download VPN Connection File

Previous Next

+10 Streak pts Mark Complete & Next

Powered by HACKTHEBOX

- Question : What system user has full control over the C:\Users directory? Answer: bob.smith

I used the command icacs against the directory C and found out the access set.

The top screenshot shows the HackTheBox Academy interface. A notification banner at the top states: "HTB CPTS Attempts Blocked Starting May 18, 2025. HTB CPTS exam lab updates may last up to 10 days. Retake users will be issued new vouchers." A green success message on the right says: "Success! Congratulations! You earned 0 cubes!" The main content area is titled "Questions" and contains the following text:

Answer the question(s) below to complete this Section and earn cubes!

Target(s): 10.129.204.75 (ACADEMY-MISC-WS01)

Life Left: 91 minute(s) + Terminate X

RDP to 10.129.204.75 (ACADEMY-MISC-WS01) with user "htb-student" and password "Academy_WinFun!"

0 What system user has full control over the c:\users directory?

bob.smith

Submit

Previous Next

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The bottom screenshot shows a Windows Remote Desktop session titled "FreeRDP: 10.129.204.75". The desktop background is blue with a white dragon logo. A Windows PowerShell window is open, displaying the following commands and output:

```
PS C:\Users\htb-student> cd ..
PS C:\Users> cd ..
PS C:\> dir

Directory: C:\

Mode                LastWriteTime         Length Name
----                -
d-----        6/23/2021 10:20 AM             75afac25577675a9efaf62485682
d-----        9/7/2020  1:43 PM             Academy
d-----       12/7/2019  1:14 AM             PerfLogs
d-----       1/31/2022  4:40 PM             Program Files
d-----       1/31/2022  3:01 PM             Program Files (x86)
d-----       1/31/2022  3:02 PM             Users
d-----       1/31/2022  4:07 PM             Windows

PS C:\> cd .\Academy\
PS C:\Academy> dir

Directory: C:\Academy

Mode                LastWriteTime         Length Name
----                -
-r-----        9/7/2020 12:17 PM             32 flag.txt

PS C:\Academy> cat .\flag.txt
afread077d3ahcd59efcfe1e413c75
PS C:\Academy> icacls c:\users
c:\users Everyone:(OI)(CI)(RX)
NT AUTHORITY\SYSTEM:(OI)(CI)(F)
BUILTIN\Administrators:(OI)(CI)(F)
WS01\bob.smith:(OI)(CI)(F)
BUILTIN\Users:(OI)(CI)(RX)

Successfully processed 1 files; failed processing 0 files
PS C:\Academy>
```

I used the command net share to list the shares and found the directory of the folder as shown

```
Windows PowerShell
PS C:\Users\hnb-student> net share

Share name Resource Remark
-----
C$ C:\ Default share
IPC$ C:\WINDOWS Remote IPC
ADMIN$ C:\Users\hnb-student\Desktop\Company Data Remote Admin
Company Data C:\Users\hnb-student\Desktop\Company Data

The command completed successfully.
PS C:\Users\hnb-student>
```

FreeRDP: 10.129.204.75

12:20:24:717 [69647:69648] [INFO][com.freerdp.channels.rdpnd.client] - [static] Loaded fake backend for rdpnd
12:20:24:721 [69647:69648] [INFO][com.freerdp.channels.drdynvc.client] - Loading Dynamic Virtual Channel rdpgfx

Failure 'self-signed certificate (18)' at stack position 0
_FORMAT_BGRX32
_FORMAT_BGRA32
Loaded fake backend for rdpnd
Dynamic Virtual Channel rdpgfx
system error 110: Connection timed out
_set_last_error_ex ERRCONNECT_CONNECT_TRANSPORT_FAILED [0x0002000D]
Failure 'self-signed certificate (18)' at stack position 0
_FORMAT_BGRX32
_FORMAT_BGRA32

Question: Identify one of the non-standard update services running on the host. Submit the full name of the service executable (not the DisplayName) as your answer. Answer: FoxitReaderUpdateService.exe

I used the Get-Wmiobject command shown below with the filters to get services running with names including “update” and got the answer

The image shows a Windows PowerShell terminal window and a HackTheBox Academy challenge page. The PowerShell window displays the output of the command `Get-WmiObject Win32_Service | Where-Object { $_.Name -like '*update*' -and $_.StartName -ne 'System' } | Select-Object Name, DisplayName, PathName`, which lists several update services, including `FoxitReaderUpdateService`.

The HackTheBox Academy page shows a challenge titled "HTB CPTS Attempts Blocked Starting May 18, 2025". The challenge description asks the user to identify one of the non-standard update services running on the host. The user has submitted the answer `FoxitReaderUpdateService.exe`, which is marked as correct. The page also shows a success message: "Success! Congratulations! You earned 0 cubes!".

Windows PowerShell terminal output:

```
PS C:\Users\htb-student> net share
Share name      Resource
-----
C$              C:\
IPC$            C:\WINDOWS
ADMIN$          C:\Users\htb-student\Desktop\Company Data
Company Data    C:\Users\htb-student\Desktop\Company Data

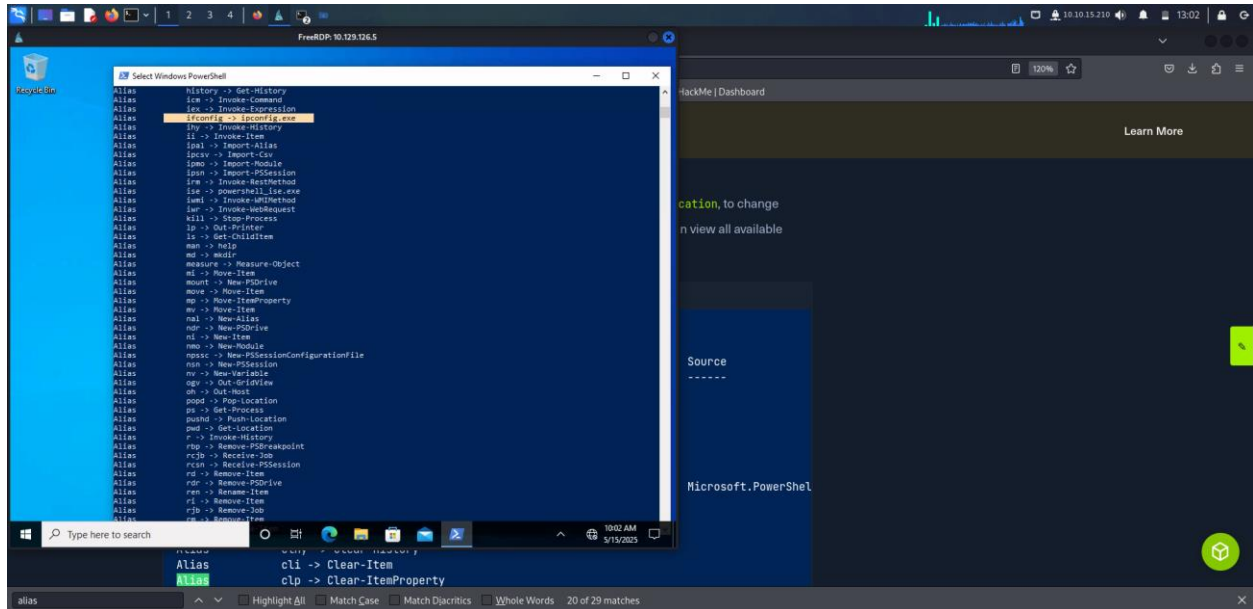
The command completed successfully.

PS C:\Users\htb-student> Get-WmiObject Win32_Service | Where-Object { $_.Name -like '*update*' -and $_.StartName -ne 'System' } | Select-Object Name, DisplayName, PathName
Name            DisplayName          PathName
-----
edgeupdate      Microsoft Edge Update Service (edgeupdate)  "C:\Program Files (x86)\Microsoft\EdgeUpdate\M...
edgeupdate      Microsoft Edge Update Service (edgeupdate)  "C:\Program Files (x86)\Microsoft\EdgeUpdate\M...
FoxitReaderUpdateService Foxit Reader Update Service "C:\Program Files (x86)\Foxit Software\Foxit Re...
```

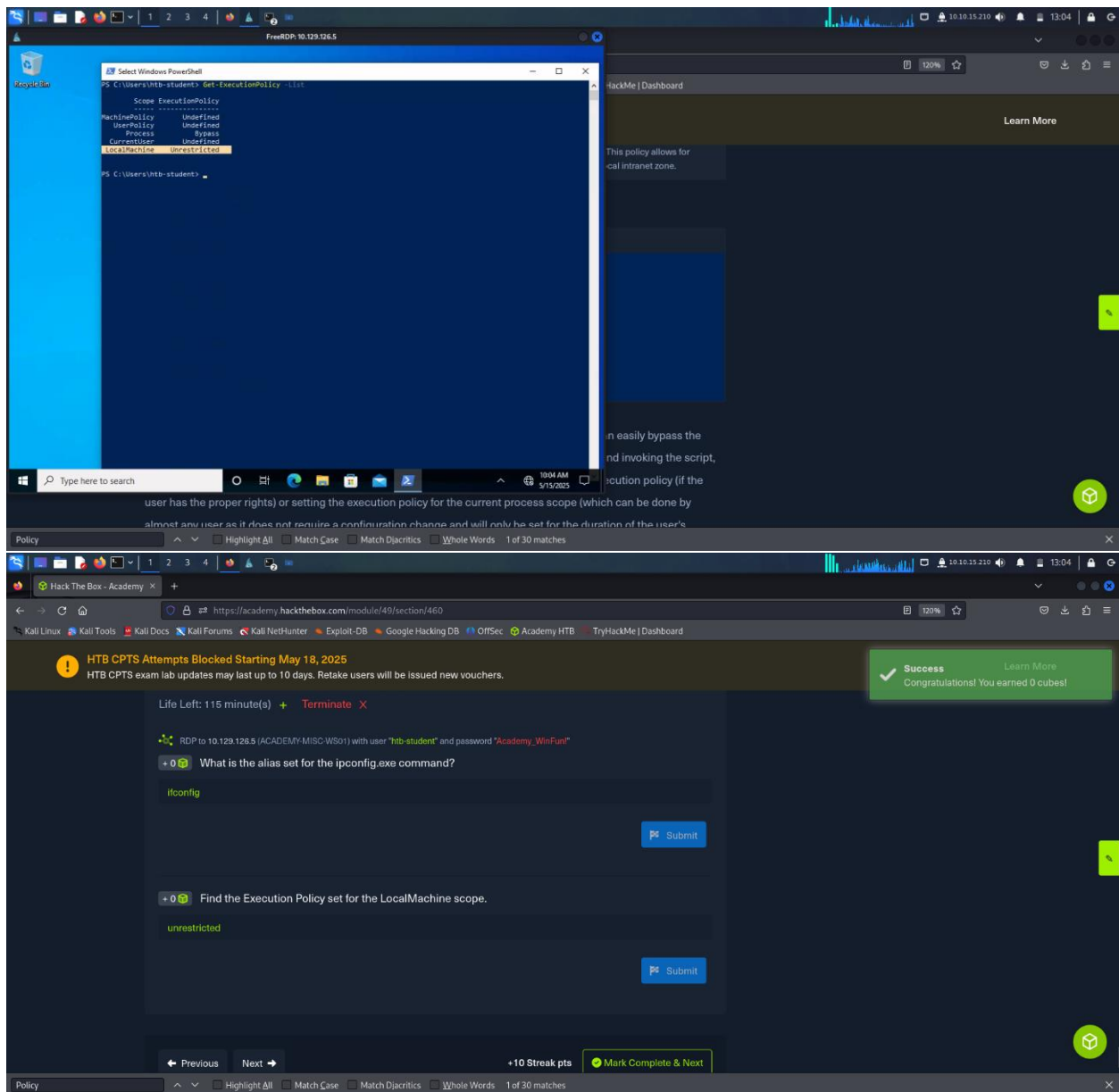
HackTheBox Academy challenge details:

- Target(s): 10.129.204.75 (ACADEMY-MISC-WS01)
- Life Left: 8 minute(s)
- Question: Identify one of the non-standard update services running on the host. Submit the full name of the service executable (not the DisplayName) as your answer.
- Submitted Answer: FoxitReaderUpdateService.exe
- Status: Success! Congratulations! You earned 0 cubes!

- Question: What is the alias set for the ipconfig.exe command? Answer: ifconfig
 - Question: Find the Execution Policy set for the LocalMachine scope. Answer: unrestricted
- I used the Get Alias command to find the alias set for ipconfig.exe .

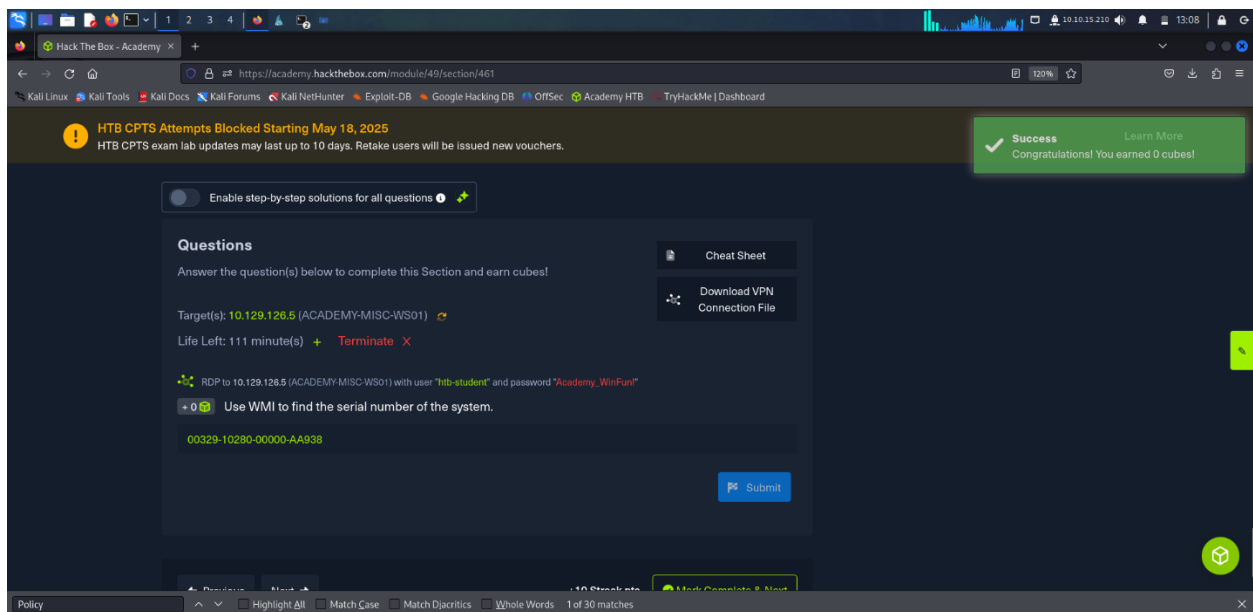
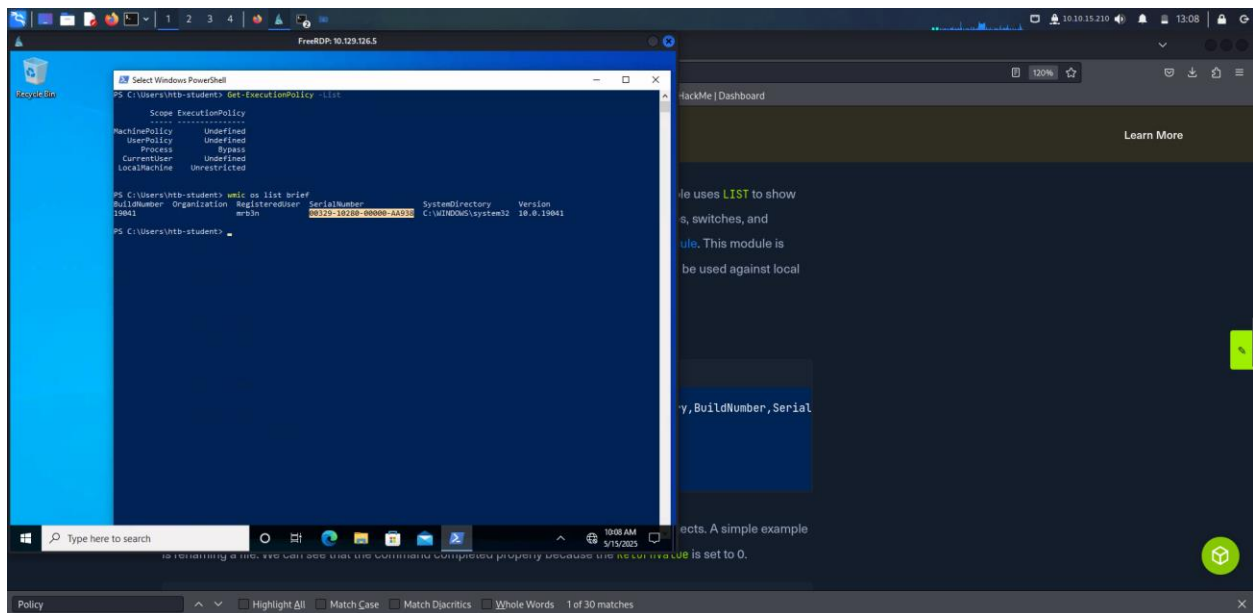


I then used the Get-ExecutionPolicy –list command to list out the workstation’s execution policies as shown below.



Question : Use WMI to find the serial number of the system. Answer: 0329c-10280-00000-AA938

I used the command "wmic os list brief" to list the serial number of the workstation as shown below



Question: Find the SID of the bob.smith user

I used the `wmic useraccount get name,sid` command to list the users with their names and SID numbers and got Bob's SID number as shown below.

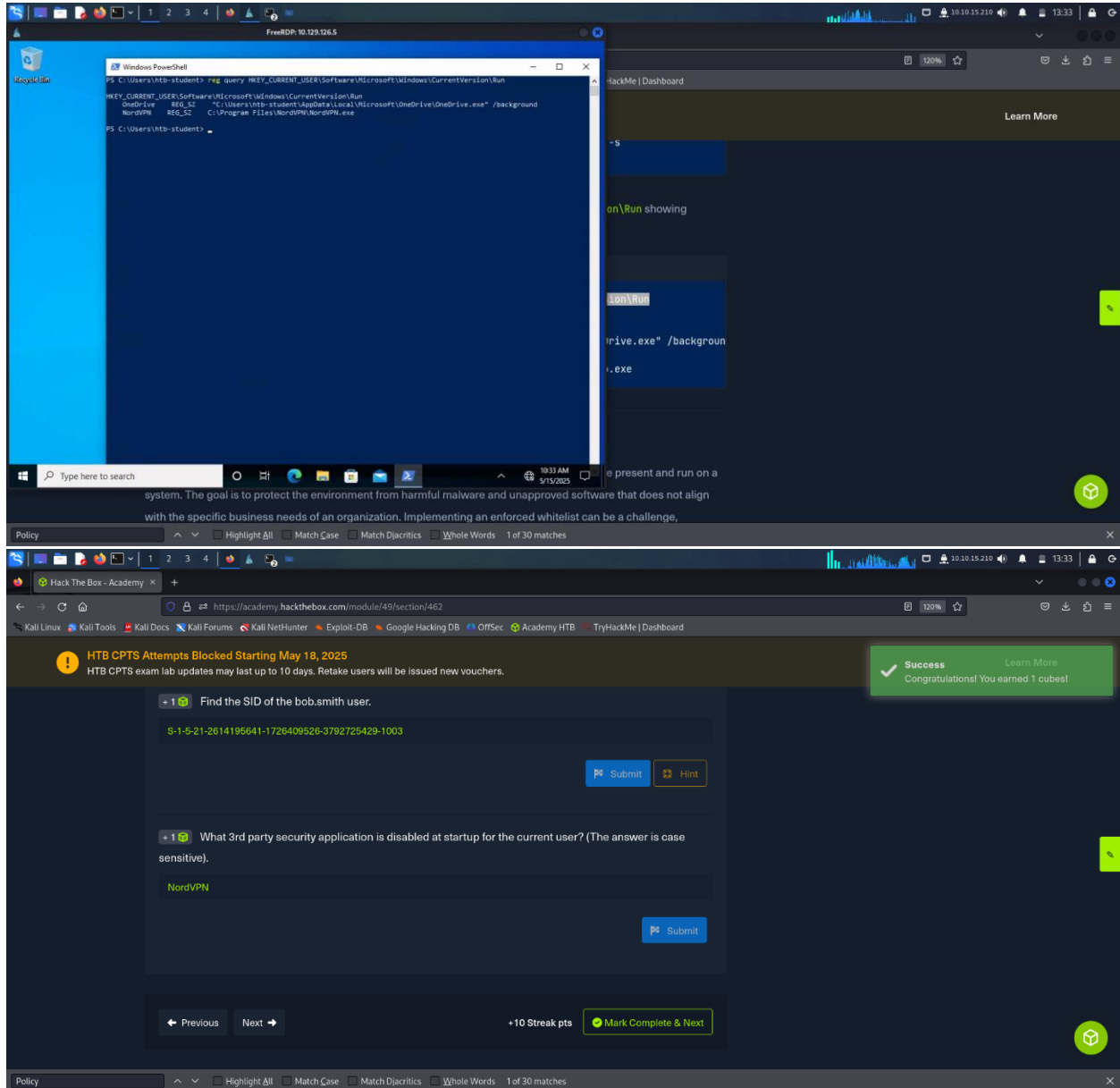
The screenshot shows a Windows PowerShell window running the command `wmic useraccount get name,sid`. The output lists several users and their SIDs. The user `bob.smith` is highlighted with a red box, showing its SID as `S-1-5-21-2614195641-1726409526-3792725429-1003`.

Below the PowerShell window, the HackMe dashboard is visible. It shows a challenge titled "Find the SID of the bob.smith user." with a target IP of `10.129.126.5` and a life left of 95 minutes. The challenge is marked as "Success" with a green checkmark and the message "Congratulations! You earned 1 cube!".

Name	SID
Administrator	S-1-5-21-2614195641-1726409526-3792725429-500
bob.smith	S-1-5-21-2614195641-1726409526-3792725429-1003
DefaultAccount	S-1-5-21-2614195641-1726409526-3792725429-503
DefaultUser0	S-1-5-21-2614195641-1726409526-3792725429-1008
Guest	S-1-5-21-2614195641-1726409526-3792725429-501
htb-student	S-1-5-21-2614195641-1726409526-3792725429-1002
wp3m	S-1-5-21-2614195641-1726409526-3792725429-1001
WMI	S-1-5-21-2614195641-1726409526-3792725429-984

Question : What 3rd party security application is disabled at startup for the current user? (The answer is case sensitive). Answer: NordVPN

I used the reg query command to list the programs disabled at startup for the current user as shown below.



- Question: List the SID associated with the user account Jim you created. Answer: S-1-5-21-2614195641-1726409526-3792725429-1006

I used the wmic useraccount get name,sid to get the SID of JIM as shown below.

The screenshot shows a Windows terminal window with the following output:

Name	SID
Administrator	S-1-5-21-2614195641-1726409526-3792725429-500
bob.smith	S-1-5-21-2614195641-1726409526-3792725429-1003
DefaultAccount	S-1-5-21-2614195641-1726409526-3792725429-360
DefaultUser0	S-1-5-21-2614195641-1726409526-3792725429-1000
Guest	S-1-5-21-2614195641-1726409526-3792725429-501
hth-student	S-1-5-21-2614195641-1726409526-3792725429-1002
Jim	S-1-5-21-2614195641-1726409526-3792725429-1006
root	S-1-5-21-2614195641-1726409526-3792725429-1004
MDAUtilityAccount	S-1-5-21-2614195641-1726409526-3792725429-504

The terminal is overlaid on a HackMe dashboard. The dashboard shows a 'Success' message: 'Congratulations! You earned 1 cubest!'. Below the message, there are two questions:

- List the SID associated with the user account Jim you created.
Answer: S-1-5-21-2614195641-1726409526-3792725429-1006
- List the SID associated with the HR security group you created.
Submit your answer here...

Question : List the SID associated with the HR security group you created. Answer: S-1-5-21-2614195641-1726409526-3792725429-1007

I used the wmic group get name,sid to get the SID of the HR group as shown below

The screenshot displays a HackMe dashboard with a Windows terminal window open. The terminal shows the following commands and output:

```
PS C:\Users\htr-student> wmic useraccount get name,sid
Name SID
Administrator S-1-5-21-2614195641-1726409526-3792725429-500
bob.smith S-1-5-21-2614195641-1726409526-3792725429-503
DefaultAccount S-1-5-21-2614195641-1726409526-3792725429-505
DefaultUser S-1-5-21-2614195641-1726409526-3792725429-1000
Guest S-1-5-21-2614195641-1726409526-3792725429-501
htr-student S-1-5-21-2614195641-1726409526-3792725429-1003
Jim S-1-5-21-2614195641-1726409526-3792725429-1006
josh S-1-5-21-2614195641-1726409526-3792725429-1001
WDAGUtilityAccount S-1-5-21-2614195641-1726409526-3792725429-504

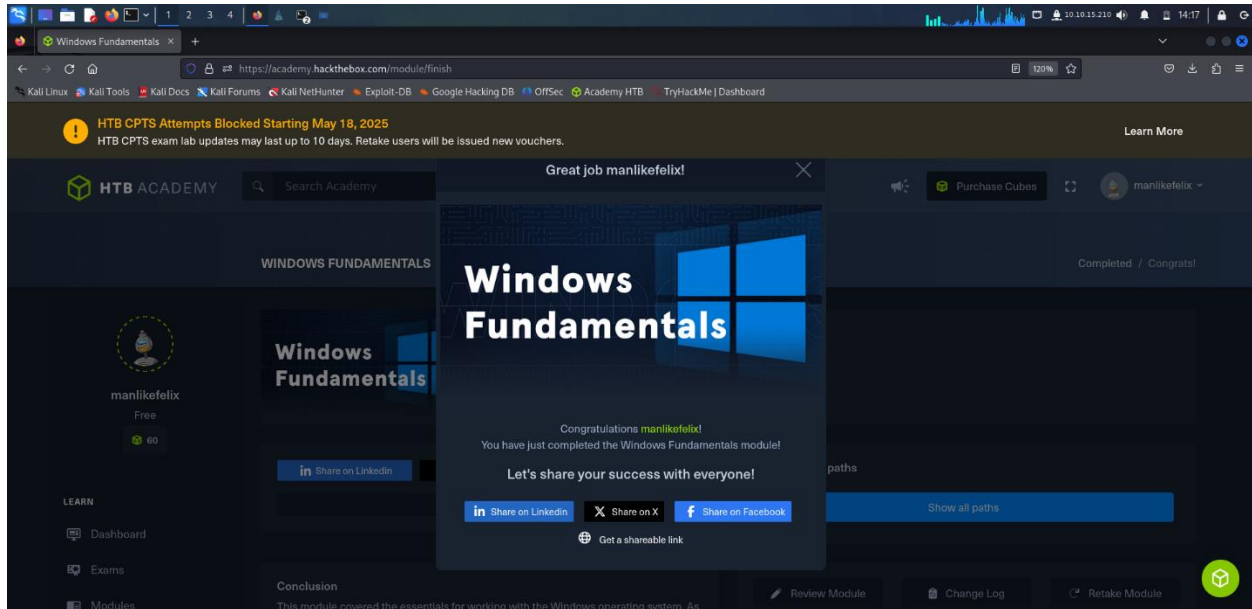
PS C:\Users\htr-student> wmic group get name,sid
Name SID
Access Control Assistance Operators S-1-5-32-579
Administrators S-1-5-32-544
Backup Operators S-1-5-32-551
Cryptographic Operators S-1-5-32-569
Distributed COM Users S-1-5-32-562
Event Log Readers S-1-5-32-573
Guests S-1-5-32-546
Hyper-V Administrators S-1-5-32-576
IIS_IUSRS S-1-5-32-568
Network Configuration Operators S-1-5-32-550
Performance Log Users S-1-5-32-559
Performance Monitor Users S-1-5-32-558
Power Users S-1-5-32-547
Remote Desktop Users S-1-5-32-555
Security Management Users S-1-5-32-560
Replicator S-1-5-32-552
System Managed Accounts Group S-1-5-32-561
Users S-1-5-32-545
```

The HackMe interface shows a challenge titled "List the SID associated with the user account Jim you created." with the answer "S-1-5-21-2614195641-1726409526-3792725429-1006". Below it, another challenge titled "List the SID associated with the HR security group you created." has the answer "S-1-5-21-2614195641-1726409526-3792725429-1007". A success message indicates "Congratulations! You earned 1 cube!".

3. Shareable Link and Final Screenshot

<https://academy.hackthebox.com/achievement/1919044/49>

- Final screenshot showing module completion.



4. Social Media Posting

<https://www.facebook.com/sharer/sharer.php?u=https://academy.hackthebox.com/achievement/1919044/49>

https://www.linkedin.com/uas/login?session_redirect=https%3A%2F%2Fwww.linkedin.com%2FshareArticle%3Furl%3Dhttps%3A%2F%2Facademy.hackthebox.com%2Fachievement%2F1919044%2F49%26title%3DCompleted%2520Windows%2520Fundamentals!%26summary%3DI%2520just%2520completed%2520the%2520module%2520Windows%2520Fundamentals%2520in%2520HTB%2520Academy!%23hackthebox%23htbacademy%23cybersecurity&source=HTBAcademy

[https://x.com/intent/post?url=https%3A%2F%2Facademy.hackthebox.com%2Fachievement%2F1919044%2F49&text=I just completed module Windows Fundamentals in HTB Academy!&hashtags=hackthebox%2Chtbacademy%2Ccybersecurity](https://x.com/intent/post?url=https%3A%2F%2Facademy.hackthebox.com%2Fachievement%2F1919044%2F49&text=I%20just%20completed%20module%20Windows%20Fundamentals%20in%20HTB%20Academy!&hashtags=hackthebox%2Chtbacademy%2Ccybersecurity)

5. Conclusion

This module provided a solid foundation for working with the Windows operating system from a security perspective. By exploring system structure, file systems, permissions, services, and security features, we have built the knowledge necessary to navigate and assess Windows environments effectively. Understanding these core components is essential not only for identifying vulnerabilities but also for leveraging Windows systems as part of a broader offensive or defensive strategy.